Aryasatya Muhammad Aqsel

38 Mugas Barat Raya Street, Mugassari, South Semarang, Semarang, Central Java 50249, Indonesia
+62 88274016649 | aryasatyaaqsel@gmail.com

EDUCATION

University of Dian Nuswantoro **Semarang, Indonesia**
Bachelor of Informatics Engineering 2022-2026

- **GPA:** 3.48 / 4.0
- **Honors:** Very Satisfactory
- **Relevant Coursework:** Machine Learning, Data Mining, Database Systems, Statistics, Computer Vision

WORK & LEADERSHIP EXPERIENCE

Department of Communications and Information Technology **Semarang, Indonesia**
Intern, Software Development Team May 2025 – Jun 2025

- Contributed to a team of 4 in developing a Laravel-based intern activity reporting system, replacing a previously manual process for tracking intern submissions
- Conducted functional testing and bug identification across system features to ensure reliability prior to deployment
- Monitored backend performance and assisted in MySQL database oversight to maintain data integrity

SKILLS, ACTIVITIES & INTERESTS

Languages: Native in Bahasa Indonesia; Conversational Proficiency in English; Basic in Javanese
Technical Skills: Python, TensorFlow/Keras, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, Git, Next.js, React, Tailwind CSS, HTML/CSS
Certifications & Training:

- Associate Data Scientist BNSP / LSP UDINUS (Apr 2026)
- AI Literacy for Everyone BASS Training Center × Google.org & ADB (Nov 2025)
- UX Writing Short Class MySkill (Oct 2024)

Interests: Formula 1 data engineering and race strategy analytics; environmental data science and causal inference

PROJECTS & RESEARCH

Jakarta Air Quality Analysis COVID-19 as a Natural Experiment | *Python, Pandas, Statsmodels, Streamlit, React*

- Analyzed 5 years of Jakarta PM2.5 data (2019-2023); Jakarta exceeds WHO annual guideline by 7.6×, with 89.8% of days above the 24-hour threshold
- Applied Interrupted Time Series regression with HAC standard errors to isolate PSBB lockdown effect: -17.4% PM2.5 reduction (95% CI, p=0.050) after controlling for weather confounders
- Built dual-frontend dashboard (Streamlit + React/Vite) with modular data pipeline from OpenAQ API and Open-Meteo ERA5

F1 Race Strategy Intelligence Dashboard | *Python, FastF1, XGBoost, SHAP, Streamlit, Next.js*

- Built dual-frontend F1 analytics platform covering 2022-2024 seasons: telemetry, tyre strategy, and driver head-to-head comparisons; deployed on Streamlit Cloud and Vercel

- Developed ML race-outcome classifier (DNF / Podium / Points / Outside Points) with SHAP explainability integrated into interactive dashboard
- Engineered modular data pipeline from FastF1 and Ergast APIs with structured feature engineering for ML and visualization

Coffee Bean Classification Using ResNet50 | *Python, TensorFlow/Keras, OpenCV*

- Developed two-stage transfer learning + fine-tuning pipeline to classify 4 Indonesian coffee bean varieties (Arabica, Robusta, Excelsa, Liberica) across 2,000 images
- Achieved 94.5% test accuracy; primary confusion source identified as Excelsa Arabica similarity (72.7% of errors), documented in analysis
- Published in JAIC (SINTA 3 indexed)